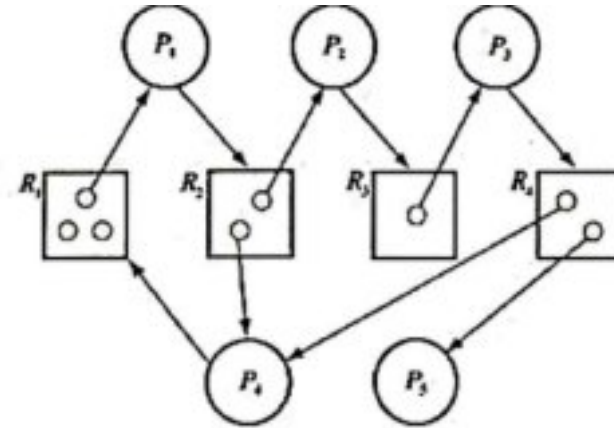


1. Consider the following resource allocation graph.



Do the following problems:

- Convert it to the matrix representation (i.e., Allocation, Request and Available).
 - Apply Banker's Algorithm to the matrix. And find if there is a deadlock? If there is a deadlock, which processes are involved? If not, share the safe sequence.
2. A hard disk system has 800 tracks where number of sectors/track is 150, number of bytes/sector is 512. The time required to move head from one track to adjacent track is 2 ms. Rotation speed is 7200 RPM.